PHYS4055 Week 9 24/9/26.

1. Want to show $\langle l_1 \| C^k \| l_2 \rangle = (-1)^{l_1} \sqrt{(2l_1+1)(2l_2+1)} \cdot \begin{pmatrix} l_1 & k & l_2 \\ 0 & 0 & 0 \end{pmatrix}$

Recall $C^k_q = \sqrt{\frac{4\pi}{2k+1}} Y^k_q$

By the Wigner-Eckart Theorem

$$\langle l_1 m_1 | Y^k_q | l_2 m_2 \rangle = (-1)^{l_1 - m_1} \begin{pmatrix} l_1 & k & l_2 \\ -m_1 & q & m_2 \end{pmatrix} \langle l_1 \| Y^k \| l_2 \rangle \quad (*)$$

LHS:

~~$\int d\theta\, d\phi\, \sin\theta\, Y_{l_1 m_1}(\theta,\phi)\, Y_{l_2 m_2}$~~

$\int d\theta\, d\phi\, \sin\theta\, Y^*_{l_1 m_1}\, Y_{kq}\, Y_{l_2 m_2}$

Note:

$$\int d\theta\, d\phi\, \sin\theta\, Y_{l_1 m_1} Y_{l_2 m_2} Y_{l_3 m_3} = \sqrt{\frac{(2l_1+1)(2l_2+1)(2l_3+1)}{4\pi}} \begin{pmatrix} l_1 & l_2 & l_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \quad (**)$$

take as given

LHS

$= \int d\theta\, d\phi\, \sin\theta\, Y^*_{l_1 m_1}\, Y_{kq}\, Y_{l_2 m_2}$ → using $Y^*_{lm} = (-1)^m Y_{l-m}$

$= (-1)^{m_1} \int d\theta\, d\phi\, \sin\theta\, Y_{l_1 -m_1}\, Y_{kq}\, Y_{l_2 m_2}$

$$= (-1)^{m_1} \sqrt{\frac{(2l_1+1)(2k+1)(2l_2+1)}{4\pi}} \begin{pmatrix} l_1 & k & l_2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & k & l_2 \\ -m_1 & q & m_2 \end{pmatrix} \quad (***)$$

Comparing $(*)$ and $(***)$, assuming 3J symbol is nonzero:

$|l_1 - k| \le l_2 \le l_1 + k$
$l_1 + k + l_2$ an integer
$-m_1 + q + m_2 = 0$
} must be satisfied

$$\therefore \langle l_1 \| Y^k \| l_2 \rangle = (-1)^{-l_1 + 2m_1} \sqrt{\frac{(2l_1+1)(2k+1)(2l_2+1)}{4\pi}} \begin{pmatrix} l_1 & k & l_2 \\ 0 & 0 & 0 \end{pmatrix}$$

Note $l \in \{0, 1, 2, \ldots\}$, $-l \le m \le l$.

$\therefore l \pm m$ is integer, $\therefore 2(l \pm m)$ is an even integer

$\therefore (-1)^{-l_1 + 2m_1} = (-1)^{l_1} \underbrace{(-1)^{-2(l_1 - m_1)}}_{=1} = (-1)^{l_1}$

$$\therefore \langle l_1 \| Y^u \| l_2 \rangle = (-1)^{l_1} \sqrt{\frac{(2l_1+1)(2u+1)(2l_2+1)}{4\pi}} \begin{pmatrix} l_1 & u & l_2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\therefore \langle l_1 \| C^u \| l_2 \rangle = (-1)^{l_1} \sqrt{(2l_1+1)(2l_2+1)} \begin{pmatrix} l_1 & k & l_2 \\ 0 & 0 & 0 \end{pmatrix}$$

2. $\underset{\sim}{u} \cdot \underset{\sim}{v} = -u_1 v_{-1}^* + u_0 v_0^* - u_{-1} v_1$

Recall, standard basis: $\underset{\sim}{A} = A_x \hat{\underset{\sim}{e}}_x + A_y \hat{\underset{\sim}{e}}_y + A_z \hat{\underset{\sim}{e}}_z$

spherical basis: $\underset{\sim}{A} = A_1 \hat{\underset{\sim}{e}}_1 + A_0 \hat{\underset{\sim}{e}}_0 + A_{-1} \hat{\underset{\sim}{e}}_{-1}$

where $\hat{\underset{\sim}{e}}_1 = -\frac{1}{\sqrt{2}} (\hat{\underset{\sim}{e}}_x + i \hat{\underset{\sim}{e}}_y)$, $\hat{\underset{\sim}{e}}_{-1} = \frac{1}{\sqrt{2}} (\hat{\underset{\sim}{e}}_x - i \hat{\underset{\sim}{e}}_y)$

$\hat{\underset{\sim}{e}}_0 = \hat{\underset{\sim}{e}}_z$

$$\underset{\sim}{V} = \sum_q V_q \hat{\underset{\sim}{e}}_q^*, \quad q = 0, \pm 1$$

Note: $\hat{\underset{\sim}{e}}_{\pm 1}^* = -\hat{\underset{\sim}{e}}_{\mp 1}$ (A), $\delta_{qq'} = \hat{\underset{\sim}{e}}_q^* \cdot \hat{\underset{\sim}{e}}_{q'}$ (B), $(\hat{\underset{\sim}{e}}_0^* = \hat{\underset{\sim}{e}}_0)$

$$\therefore \underset{\sim}{u} \cdot \underset{\sim}{v} = (u_1 \hat{\underset{\sim}{e}}_1^* + u_0 \hat{\underset{\sim}{e}}_0^* + u_{-1} \hat{\underset{\sim}{e}}_{-1}^*) \cdot (v_1 \hat{\underset{\sim}{e}}_1^* + v_0 \hat{\underset{\sim}{e}}_0^* + v_{-1} \hat{\underset{\sim}{e}}_{-1}^*)$$

$$= (u_1 \hat{\underset{\sim}{e}}_{-1} + u_0 \hat{\underset{\sim}{e}}_0 + u_{-1} \hat{\underset{\sim}{e}}_1) \cdot (-v_1 \hat{\underset{\sim}{e}}_{-1} + v_0 \hat{\underset{\sim}{e}}_0 - v_{-1} \hat{\underset{\sim}{e}}_1) \quad \text{use (A)}$$

$$= -u_1 v_{-1} + u_0 v_0 - u_{-1} v_1 \quad \text{use (B)}$$

$$= \sum_q (-1)^q u_q v_{-q}$$

3. Lecture exercise:



$$\langle IJFM | (T^u \cdot U^u) | I'J'F'M' \rangle = (-1)^{I'+J+F} \langle I \| T^u \| I' \rangle \langle J \| U^u \| J' \rangle \begin{Bmatrix} I & J & F \\ J' & I' & k \end{Bmatrix}$$

Hints 1. Use $\mathbb{I} = \sum_{\gamma'' J'' M''} |\gamma'' J'' M'' \rangle \langle \gamma'' J'' M''|$

2. Use Wigner-Eckart theorem



Want to show:

$$\langle J_1 J_2 JM | \underset{\sim}{J}_1 \cdot \underset{\sim}{J}_2 | J_1' J_2' JM \rangle = \tfrac{1}{2} [J(J+1) - J_1(J_1+1) - J_2(J_2+1)]$$

$\underset{\sim}{J}_1$ acts only only J_1, $\underset{\sim}{J}_2$ similarly on J_2, J_2'

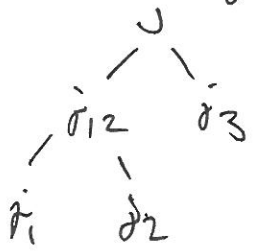
$$\langle J_1 J_2 JM | \underset{\sim}{J}_1 \cdot \underset{\sim}{J}_2 | J_1' J_2' JM \rangle$$
$$= (-1)^{J_1' + J_2 + J} \langle J_1 \| J_1 \| J_1' \rangle \langle J_2 \| J_2 \| J_2' \rangle \begin{Bmatrix} J_1 & J_2 & J \\ J_2' & J_1' & 1 \end{Bmatrix}$$

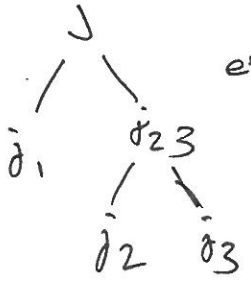
Use: $\langle J \| J \| J' \rangle = \delta_{JJ'} \sqrt{J(J+1)(2J+1)}$

↳ exercise (Hint: use Wigner, Eckart)

$$\begin{Bmatrix} J_1 & J_2 & J \\ J_2' & J_1' & 1 \end{Bmatrix} = \frac{(-1)^{J_2 + J + J_1 + 1}}{\sqrt{J_1(J_1+1)(2J_1+1)}} \frac{J_1(J_1+1) + J_2(J_2+1) - J(J+1)}{2\sqrt{J_2(J_2+1)(2J_2+1)}}$$

↑ take as given





etc. } coupling schemes of three angular angular momenta

6J symbol ⟷ with going between schemes

$$\therefore \langle J_1 J_2 JM | \underset{\sim}{J}_1 \cdot \underset{\sim}{J}_2 | J_1' J_2' JM \rangle$$
$$= \underbrace{(-1)^{J_1 + J_2 + J} (-1)^{J_2 + J + J_1 + 1}}_{\text{simplifies to } (-1)^1 \text{ by phase factor rules}} \tfrac{1}{2} \left(J_1(J_1+1) + J_2(J_2+1) - J(J+1) \right)$$
$$= \tfrac{1}{2} \left(J(J+1) - J_1(J_1+1) - J_2(J_2+1) \right)$$

Could have also shown this using:

$$\underset{\sim}{J} = \underset{\sim}{J}_1 + \underset{\sim}{J}_2$$
$$\underset{\sim}{J}^2 = \underset{\sim}{J}_1^2 + \underset{\sim}{J}_2^2 + 2 \underset{\sim}{J}_1 + \underset{\sim}{J}_2$$
$$\therefore \underset{\sim}{J}_1 \cdot \underset{\sim}{J}_2 = \tfrac{1}{2} \left(\underset{\sim}{J}^2 - \underset{\sim}{J}_1^2 - \underset{\sim}{J}_2^2 \right) \text{ and acting on } |J_1 J_2 JM\rangle$$

⋮